

to India's ambitious target of solar power generation reducing e-waste is a limited option and hence the 3 R's can be renamed as recycle, reuse, and regenerate.

Solar PV mass estimation

The global market is dominated by crystalline silicon (c-Si) PV modules with almost 98% market share with the rest being thin-film modules. India is virtually 100% c-Si market. Similarly, in comparison to the solar PV modules the flat plate collectors that also use glass and aluminium are insignificant. Hence the focus for end of life management will be focussed on the c-Si PV modules.

The typical material composition of a c-Si PV module, excluding the aluminium frame, is as shown in Fig. 2.

A solar PV module typically consists of glass, aluminium, copper, silver, encapsulants, semiconductor, and precious metals that can be successfully recovered and reused at the end of their useful life. By mass, today's typical crystalline silicon PV panels consist of approximately 70%

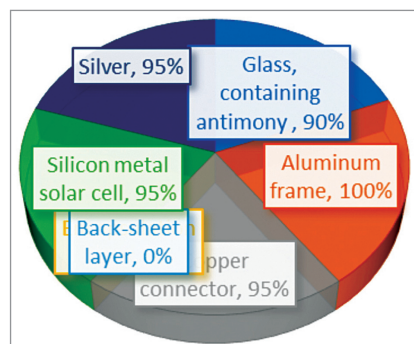


Fig. 3: Recovery percentage of various solar PV materials

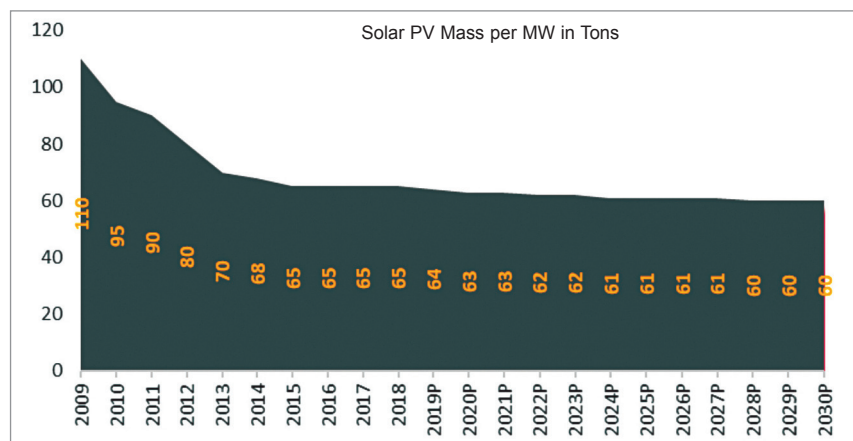


Fig. 4: Weight to power ratio of solar PV module over time

Break-up of Materials that can be Recovered from a 1,000 kg Solar PV Module Waste post end of Life Management

Table 1

Item	Quantity	Unit	% by net weight	Rate recovered (%)	Recovered material
Glass, containing antimony (0.01-1 %/kg of glass)	700	kg	70%	90	Primary white glass
Aluminum frame	180	kg	18%	100	Primary aluminum
Copper connector	10	kg	1%	95	Primary copper
Polymer-based adhesive (EVA) encapsulation layer	51	kg	5.1%	0	Diesel fuel (pyrolysis)
Back-sheet layer (based on polyvinyl fluoride)	15	kg	1.5%	0	
Silicon metal solar cell	36.5	kg	3.56%	95	Metal grade silicon
Silver	0.53	kg	0.053%	95	Primary silver
Aluminum, internal conductor	5.3	kg	0.53%	100	Primary aluminum
Copper, internal conductor	1.14	kg	1.14%	95	Primary copper
Various metal (tin, lead)	0.53	kg	0.053%	90	Primary metal
Total	1,000	kg	100%	--	

Source: Joint Research Commission, EU 2016 report

glass, 10% aluminium, 10% polymer (encapsulant and back sheet foil), 3.5% silicon semiconductor, 1% copper, and others including silver (less than 0.1%) and tin.

A report by the Joint Research Centre of European Commission gives break-up of the waste materials that can be recycled per 1,000kg of PV waste as shown in Table 1.

The percentage of material that can be recovered is as shown in Fig. 3.

The mass (weight to power ratio) of c-Si PV module is reducing with time as shown in Fig. 4. This is due to further developments in PV technologies such as material savings and improved solar cell efficiencies. This can alter the equation in Table 1 in future.

Hazardous materials

Barring glass, aluminium, and precious metals a typical c-Si solar PV module contains some of the potentially hazardous materials like polymer and antimony in the glass. At the end of life these can have negative environment and health impacts, although the situation is not alarming. The polymer material includes back-seat and encapsulant made of ethyl vinyl acetate and poly vinyl fluoride. This material cannot be recovered and is usually incinerated. However, using thermal processing at 500°C, it can be safely evaporated.

When glass is exposed to ultra-violet radiation or sunlight, antimony would leach from the glass when the waste panels get exposed to wet conditions. In March 2019, the Ministry of New and Renewable Energy had made public a draft of the blueprint for its environmentally safe handling, recycling, or disposal.

PV module waste projection by 2050

Usually, modules are warranted for 25 to 30 years and can last even longer. But as the solar industry has grown, the market has been flooded with cheaply made modules that can break down in as few as five years, which results in early loss. The PV module